DEALER BEST PRACTICE SERIES

Component Management (Strategic)

Application | Maintenance | Site Management | Component Rebuild | **Component Life Management** | Safety | MARC Management

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1.0 Introduction

The top-level goal of most mining operations is to minimize cost per ton. Maintenance and repair of machine components have a significant impact, as much as 50 to 60%, on the overall costs of supporting large mining equipment. As such, major components are key cost drivers. From a component maintenance and repair perspective, cost per ton can be influenced in several ways including:

- ✓ Maintenance & repair cost minimization … the cost element in the component cost per hour calculation,
- ✓ Life maximization … the time element in the component cost per hour calculation,
- ✓ Functional integrity … when components and systems are operating properly, they contribute positively to the productivity (Tons per Hour) of equipment.

Optimizing each of these three factors is a constant challenge for the equipment maintenance management team.

The cost ingredient can be influenced as follows:

Before versus after-failure replacement … most component replacement costs can be minimized via a planned, before-failure replacement approach.

- ✓ Component-related manpower labor … the Equipment Manager may not be able to control labor rates but he can have a significant influence on his overall labor costs by improvements in repair labor efficiency and effectiveness.
- ✓ Component inventory carrying costs … insuring that component requirements are adequate to cover short-term (protective) as well as medium/long-term (Planned Component Replacement) projections, that they do not exceed what is required, and that they are kept in a manner that they are ready to use when required.

Component lives vary substantially for mine to mine for a given piece of equipment based primarily on application severity, operating practices and maintenance effectiveness. Thus a detailed Component Management strategy and a plan to execute that strategy that takes into account not only maintenance but is also flexible enough to accommodate changes in application severity and operating practices, and assesses their potential impact on component health are critical to the long-term success of any mining support operation.

To a lesser degree, component exchange can impact downtime (availability). Because even in the extreme case, i.e. lower than expected component lives and inefficient replacement procedures, component replacement occurs relatively infrequently, the impact on availability would be no more than 6% relative to a target of <1%. However, component repairs can have a significant negative influence on machine availability (excessive machine downtime). If frequent or chronic (repetitive) component repairs are found to be the case, please refer to the guidelines found under Repair Management.

Component Management is a far-reaching process in terms of the other processes it interacts with. Clearly the success of Component Management is based upon an integrated approach that communicates directly with Condition Monitoring, Planning & Scheduling, Preventive Maintenance, and Repair Management and indirectly with Parts Management, Human Resources & Training, Performance Evaluation, and Continuous Improvement. In addition to those “internal”, on-site processes, Component Management also relies heavily on external resources such as parts and service organizations, i.e. Component Rebuild Centers and the central parts department.

This Best Practice relates specifically to the Component Management process as applied to large Off Highway Trucks and is covered under version II of Site Assessment items 2.3.1 - 2.3.5 as well as module #7 of the Gap Analysis Tool.

2.0 Implementation Steps

Implementation of an on-site Component Management process requires the development of a long-term strategy focused on achieving desired, site-specific goals. The strategy includes the definition and documentation of a logical sequence of steps and a plan to execute the strategy as follows:

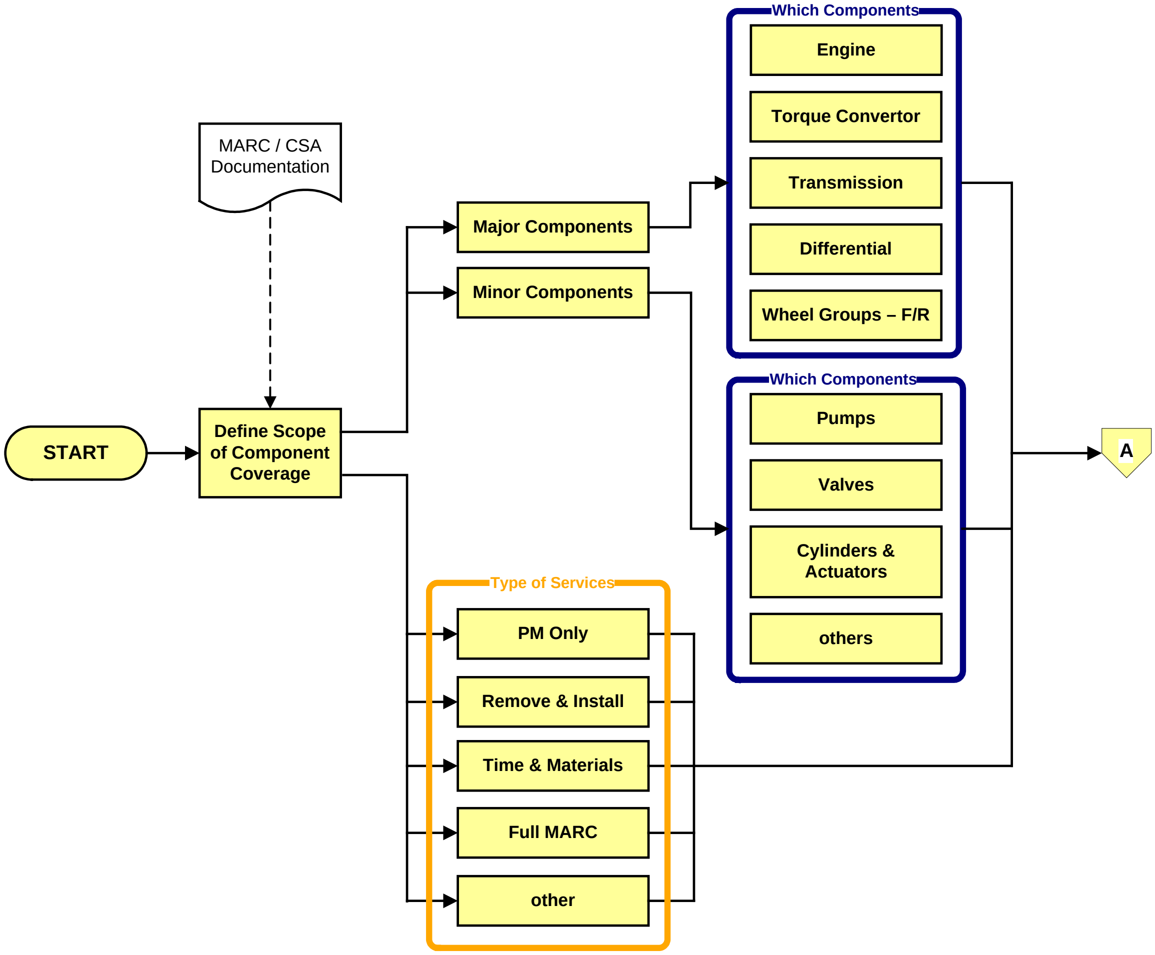

Figure #1 – Determine Scope of Component Management Coverage

- <u>Scope of Coverage</u> … *determine the extent of the operation*
  - ✓ Equipment Management Responsibility(s):
    - ➢ Equipment model(s)
    - ➢ Quantity of equipment
    - ➢ Asset Utilization (hours per machine per month)

- ✓ Type of Service:
  - ➢ Preventive Maintenance only
  - ➢ Component Removal and Installation
  - ➢ Time and Materials only
  - ➢ Full Maintenance and Repair Contract (MARC)
  - ➢ other(s)
- ✓ Component Responsibility(s):
  - ➢ Major Components:
    - Engine
    - Torque Converter
    - Transmission
    - Differential
    - Wheel Groups (front and/or rear)
    - other(s)
  - ➢ Minor Components:
    - Pumps
    - Valves
    - Cylinders and Actuators
    - other(s)

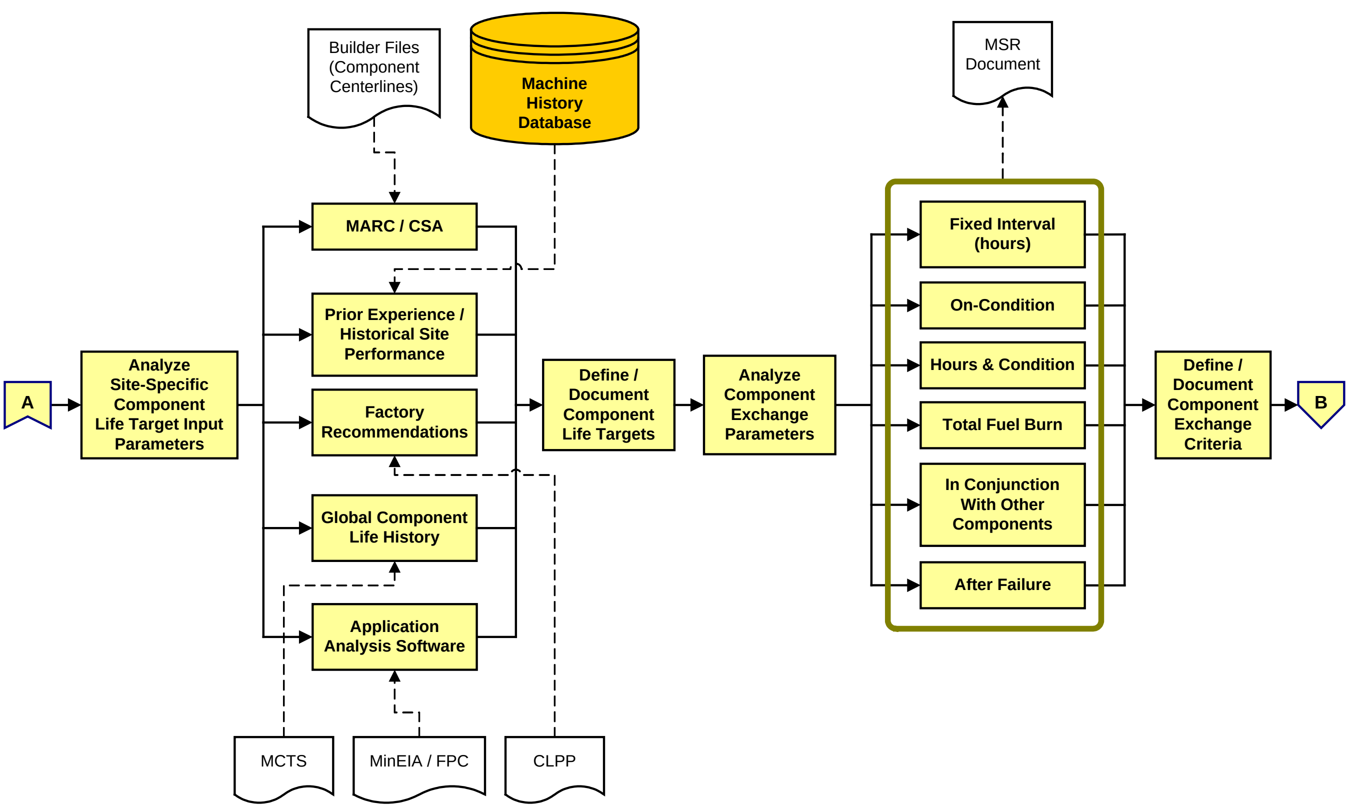

Figure #2 – Establish Goals, Objectives and Exchange Parameters for Component Management

- Goals and Objectives … *define life/cost target criteria*

- ✓ Component Life Targets:
  - ➢ Prior experience/historical site performance
  - ➢ MARC/CSA commitments (Builder files)
  - ➢ Factory recommendations (e.g. CLiPP)
  - ➢ Global component life history (e.g. MCTS)
  - ➢ Application analysis software (e.g. MinEIA, FPC, FPO, etc.)
  - ➢ other(s)
- ✓ Component Cost Targets:
  - ➢ Prior experience/historical site performance
  - ➢ MARC/CSA calculations (CSA Suite files)
  - ➢ CRC/Rebuild Center input
  - ➢ other(s)

- <u>Component Exchange Parameters</u> … *define component exchange criteria*
  - ✓ Fixed interval (hours-based)
  - ✓ On-Condition
  - ✓ Hours and Condition
  - ✓ Total Fuel Burn
  - ✓ In conjunction with other components
  - ✓ After failure
  - ✓ other(s)

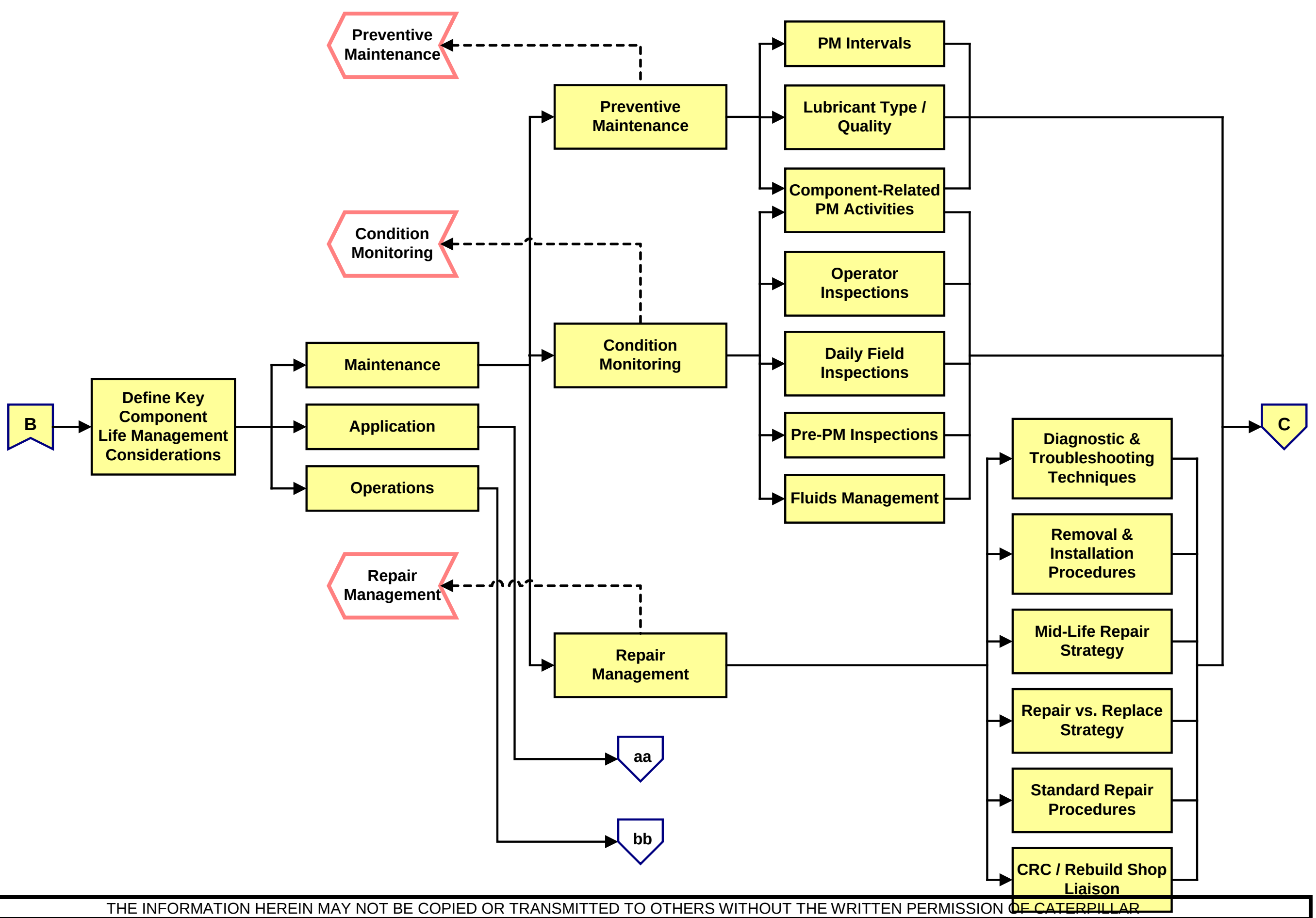

Figure #3 – Define Key Component Life Management Considerations & Acceptability Criteria

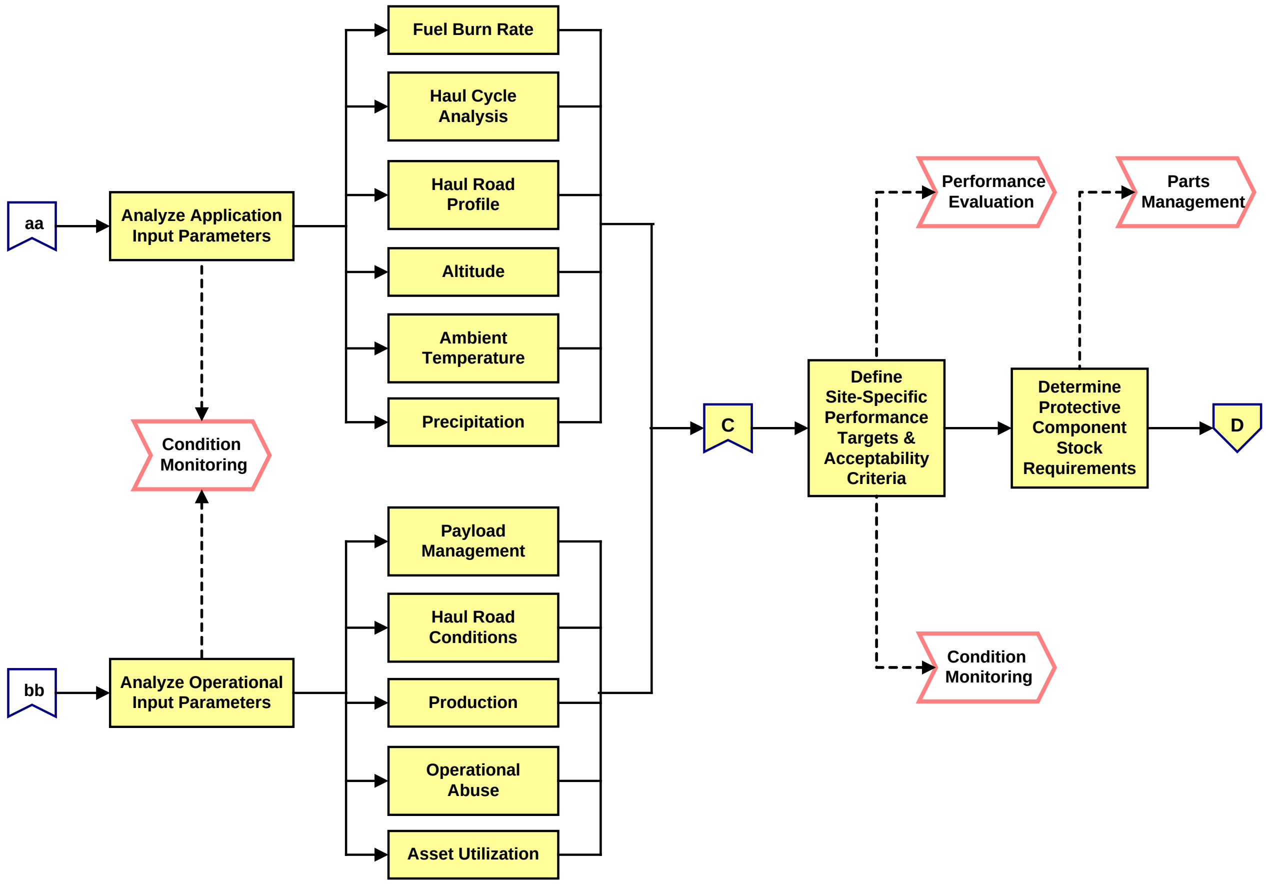

Figure #3a – Define Key Component Life Management Considerations & Acceptability Criteria (cont'd.)

- <u>Component Life Considerations</u> … *identify factors affecting component lives*
  - ✓ Maintenance
    - ➢ Preventive Maintenance:
      - PM intervals
      - Lubricant type/quality
      - Component-related PM activity(s)
    - ➢ Condition Monitoring:
      - Operator inspections
      - Periodic field inspections
      - pre-PM inspections
      - Fluids management … SOS, magnetic plug inspection, LPD, contamination control, fluids consumption, etc.
    - ➢ Repair Management:
      - Diagnostic & troubleshooting techniques
      - Removal & installation procedures
      - Mid-life reconditioning plan
      - Repair versus replacement guidelines
      - Standard repair procedures

        - CRC/rebuild shop quality

- ✓ Application:
    - ➢ Fuel burn rate
    - ➢ Haul cycle analysis
    - ➢ Haul road profiles
    - ➢ Altitude
    - ➢ Ambient temperature
    - ➢ Precipitation
    - ➢ other(s)
- ✓ Operation:
    - ➢ Payload management
    - ➢ Haul road conditions
    - ➢ Production
    - ➢ Operational abuse
    - ➢ Asset Utilization
    - ➢ other(s)

- Acceptability Criteria … *define site-specific performance targets*
    - ✓ Maintenance parameters:
        - ➢ Preventive Maintenance
        - ➢ Condition Monitoring
        - ➢ Repair Management
    - ✓ Application parameters
    - ✓ Operational parameters

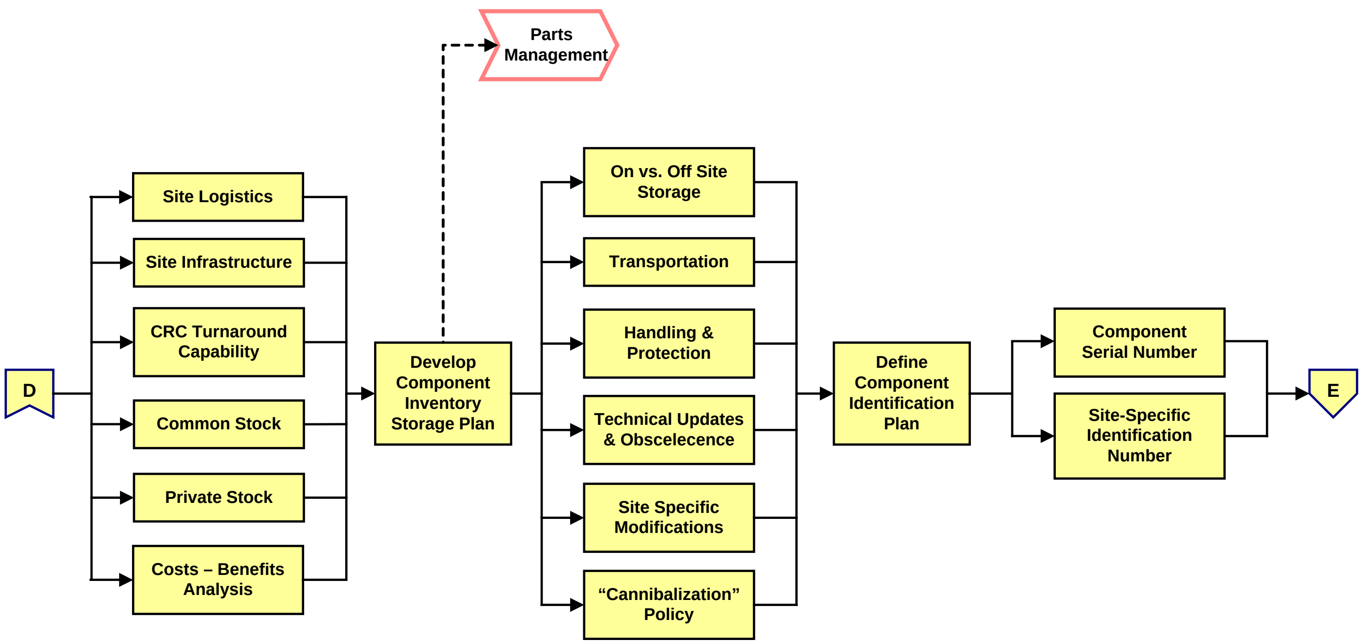

Figure #4 – Define Component Support & Management Requirements

- Component Support … *determine site-specific component requirements*

- ✓ Protective component inventory:
  - ➢ Dedicated vs. Shared stocking plan
  - ➢ Protective stock quantities (by component)
  - ➢ Site logistics
  - ➢ Site infrastructure
  - ➢ CRC/Rebuild Center turnaround capabilities
  - ➢ Cost - Benefits analyses
  - ➢ Other(s)
- ✓ PCR (Planned Component Replacement) inventory:
  - ➢ Rebuild/Reman/New component replacement strategy
  - ➢ PCR stock quantities (by component)
  - ➢ PCR stock timing (short/medium/long-term forecast) linked to Asset Utilization
  - ➢ Dedicated vs. Shared stocking plan
  - ➢ CRC/Rebuild Center turnaround capabilities
  - ➢ Other(s)
- ✓ Component Inventory Handling & Storage:
  - ➢ Handling & protection
  - ➢ Transportation
  - ➢ On vs. off-site storage
  - ➢ Reman core returns (see Parts Management)
  - ➢ Obscelecence/technical updates
  - ➢ Site-specific modifications
  - ➢ Removal/installation parts kits
  - ➢ * "Cannibalization" policy
  - ➢ Other(s)

**_NOTE_:* "Cannibalization" is the practice of taking parts from a downed machine to support the repair of a defect on another machine. This occurs either when the parts support process is weak, i.e. the quality of the on-site inventory is substandard resulting in a poor service fill level, or when an inordinate amount of unscheduled repairs take place, i.e. other maintenance and repair processes are weak resulting in a significant percentage of unplanned repair activities. In effect, "cannibalization" is a poor practice put in place to prop up one or more other weak processes.

While Global Mining neither endorses nor encourages "cannibalization", we recognize that it is an accepted practice at some sites. Thus, mine management must define a clear policy related to "cannibalization", that is, "will we do it?" and, if so, "how will we manage it?" to minimize its detrimental impact on Planning & Scheduling (and thus the timely execution of needed repairs).

- <u>Component Identification</u> … *define identification plan*
  - ✓ Component serial numbers
  - ✓ Local identifier
  - ✓ Tags/indelible marker
  - ✓ "chip"
- <u>Component Tracking</u> … *monitor component life/repair history*
  - ✓ Tracking system:

    - ➢ Manual system (paper)
    - ➢ Locally-developed system (e.g. Excel spreadsheets)
    - ➢ Computerized Maintenance Management System (third party)
    - ➢ MCTS (Caterpillar Major Component Tracking System)
    - ➢ Other(s)
  - ✓ Statistical tracking parameters:
    - ➢ Operation (operated hours, operational abuse, …)
    - ➢ Application (cumulative fuel burn, distance traveled, …)
    - ➢ Movement history
    - ➢ Failure mode analysis
    - ➢ Repair/rebuild history
    - ➢ Workorder history
    - ➢ Other(s)

- Component Commissioning … *document/validate performance baselines*
  - ✓ Determine components to be evaluated,
  - ✓ Define commissioning test procedures, forms and checklists,
  - ✓ Establish interpretation and acceptability criteria,
  - ✓ Develop and document "decision tree" guidelines for non-compliance with acceptability criteria.

- Manpower

  Determine human resources required to implement and execute the Component Management plan, i.e. define and document quantities, roles & responsibilities, personal attributes, skills, etc. (Also see Resource Requirements/Human Resources in section 5 .0).

- Training

  How can personnel competency be assessed? What training will be required to develop the skills required to perform the tasks related to Component Management?

- Infrastructure and Resource Requirements

  What infrastructure and resources will be required to execute the Component Management plan?

  - ✓ Shop work bays … appropriately equipped to handle large, heavy components
  - ✓ Specialized tooling and instrumentation
  - ✓ Machine-specific inspection and Condition Monitoring checklists
  - ✓ R&I procedures … well-defined; documented in writing
  - ✓ Data management equipment
  - ✓ External resources
  - ✓ Other(s)

  (Also see Resource Requirements/Logistical Resources in section 5 .0).

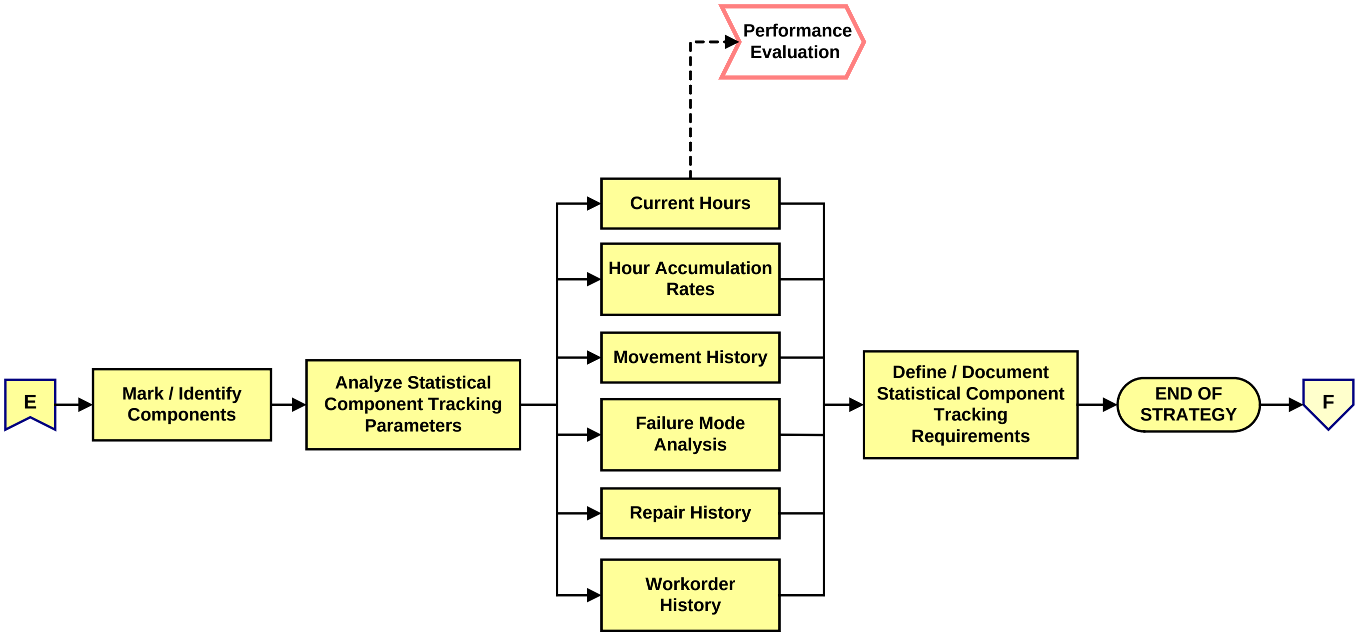

Figure #5 – Define Component Tracking & Reporting Requirements

- **Performance Measurement/Reporting**
  What defines success in Component Management? What performance measures are available that aid in quantifying our efforts in Component Management? The document "Metrics (KPI's) to Assess Process Performance" (attached) contains a complete listing and explanation of recommended performance metrics for the Component Management process.

  Once these questions have been answered, the following issues need to be taken into account when developing a plan to quantify Component Management performance:

  - ✓ Raw input data requirements,
  - ✓ Data sources,
  - ✓ Data collection frequency and responsibilities,
  - ✓ Data flow and management,
  - ✓ Calculation methodology,
  - ✓ Performance goals and acceptability criteria,
  - ✓ Analysis and interpretation roles & responsibilities,
  - ✓ Analysis and interpretation guidelines,
  - ✓ Information management/archiving,
  - ✓ Report formats (templates),
  - ✓ Report timing and frequency,
  - ✓ Communications/distribution network,
  - ✓ Distribution format and media.

3.0 Best Practice Description

Once the scope, goals/objectives, strategy and plan related to Component Management have been defined and documented, it is time to initiate the process. This Component Management Best Practice describes and focuses on the various scenarios of Component Management as one of the active elements in our maintenance and repair model, including:

- ✓ Preventive Maintenance
- ✓ Condition Monitoring
- ✓ Unscheduled Component Repair
- ✓ Component Replacement

The following is a brief description of each of those Component Management scenarios:

- **Preventive Maintenance**
  Preventive Maintenance provides a regular and convenient "window of opportunity" to perform routine service activities (e.g. change fluids and filters, perform Condition Monitoring procedures on those components and systems to monitor changes/deterioration in their performance over time). Thus the primary goals of PM as it relates to Component Management are to maintain components in peak operating condition and to assess component health and performance as compared to manufacturers' specifications and/or predetermined, site-specific acceptability criteria.

  The PM process is initiated and Component Management-related PM activities are driven by the Planning & Scheduling process using the intervals and checklists defined during the development of the Component Management strategy. Aside from the component/system service activities, the key elements of Component Management that take place during PM are Condition Monitoring routines. These routines vary from the most basic (i.e. operator interview and visual inspections), and continue through to fluids management, (e.g. SOS, filter/breather/strainer inspection), operational and special instrumented tests, and on-board electronic systems downloads which enable the site to monitor not only changes in system performance but also changes in the operation and/or application severity. Any of these changes may dictate modifications to the Component Management plan and/or the component replacement program.

  Once the PM process is completed, Component Management responsibilities continue with the documentation of results and the communication of those results to the appropriate party (typically the Fleet Analyst). All component/system condition parameters are recorded and documented in the component life/repair history with any discrepancies (condition-based exceptions) being communicated on an urgent basis and, if it is determined that action is required beyond that normally performed during PM, the equipment is typically moved to the main workshop for repair under the guidelines of Repair Management as defined in the "Component Life Considerations" topic of the "Implementation Steps" section of this Best Practice.

  The description above should make it clear that Component Management is not a stand-alone process and that success in this area requires strong linkage and integration with several other processes, most notably Planning & Scheduling, Condition Monitoring, and Repair Management.

- **Condition Monitoring**

  In addition to those activities associated with PM, component/system Condition Monitoring procedures occur in conjunction with other phases of equipment management operations, specifically during operator inspections, daily/field inspections and pre-PM inspections. The scope of coverage in each of those areas varies significantly in terms of the activities performed. A brief description of each of those areas follows:

  - ✓ Operator Inspections:

    Operators spend more time with and as a result should be more familiar with the equipment than anyone involved in its management. The scope of the operators' involvement in component condition monitoring should include both the pre-shift visual inspection as well as the post-shift component operation and performance assessment. These activities are relative low-tech when compared to other condition monitoring routines.

    The pre-shift inspection includes:

    - ➢ A visual, walk-around inspection
    - ➢ The documentation of findings on pre-shift inspection checklist or machine logbook
    - ➢ Reporting urgent defects to Dispatch or in-pit Supervision

    The post-shift routine includes:

    - ➢ Exception-based assessment of machine condition and performance
    - ➢ Recording machine performance observations on post-shift inspection checklist or machine logbook
    - ➢ Communication of observations to Operator for upcoming shift

  Please see "Operator Inspections" routine in the *Condition Monitoring* Strategic Best Practice for additional details.

  - ✓ Daily/Field Inspections:

    Daily inspections should be scheduled during convenient "windows of opportunity" such as shift changes and/or daily lubrication and refueling stops. These inspections should be performed by a highly skilled, detail-oriented Inspector, are typically quick, easy to perform (15 - 20 minutes in duration), and, once again tend to be relatively lo-tech, subjective observations although activities such as tire pressure checks and fluid additions are frequently included. The daily inspection also includes an Operator interview, a review of available information (e.g. Operators' logbook, backlog repair list, top problems summary, etc.) in addition to the visual inspection.

    Critical to the success of the daily inspection are well-guided, machine-specific inspection checklists, well-documented procedures, minimal reliance on verbal communications and the Inspectors' ability to analyze, interpret and translate Operator input into technical feedback and action such as backlog repair requests.

  - ✓ Pre-PM Inspections:

    The pre-PM inspection is performed to verify that the list of activities planned for Preventive Maintenance is accurate and complete. In order to be effective, the pre-PM inspection must be scheduled and executed in time to allow Planning to respond to feedback from the inspection. Execution of the pre-PM inspection is the responsibility of the Field Inspectors and Technicians. In addition to all of the activities normally performed during the daily field inspection, fluid sampling, magnetic plug inspections, VIMS and other on-board electronic system downloads, and backlog repair validation are included in the pre-PM inspection.

The pre-PM inspection should be accomplished during the same “windows of opportunity” as the daily field inspection and in the same amount of time (15 - 20 minutes). The same factors that drive success of the daily field inspection are also critical to the success of the pre-PM inspection.

It is important to note that for any of these activities to add value, it is imperative that accurate, complete feedback is documented and communicated in a timely fashion to the Fleet Analyst. He or she is then responsible for the analyzing the results, updating the machine/component history files and component “Hot Sheet”, and communicating advice to Planning regarding the need for repairs outside the normal plan. Only then can abnormal conditions/performance and changes in application severity be detected in time to take preventive or corrective actions that can preserve and maintain component health. This is particularly important as components approach the final stages of their predicted useful life and will, in some cases, enable component lives to be extended beyond their planned replacement interval.

- <u>Unscheduled Component Repair</u>

  Component replacement may be necessary as a result of findings that result from either the Preventive Maintenance or Condition Monitoring routines as discussed in the previous two topics. In addition, events that take place during normal equipment operations may lead to component replacement. Unexpected component problems and failures will occur at even the best-managed sites. As such, one of the goals of component management is to make the replacement as transparent as possible in terms of the downtime and costs associated with the repair or replacement.

  When an operator determines that conditions justify stopping a piece of equipment, the first order of business is to advise Dispatch of the situation. Dispatch must then send the field service team to the machine for attention. Based on the information available (from Dispatch or Operations), the field service team should collect any resources it feels are required before going to the machine. Once the field service team arrives at the machine, it uses those resources as well as operator input, field inspection, and on-board system analysis to make a determination as to whether or not the call is valid. If the call is not valid, the machine is returned to service and a field shift report is generated. This information may be used later by the Training Department to define operator training requirements.

  If the call is valid, a workorder is opened and the field service team must next determine if the problem can be backlogged for future repair or if the repair is more urgent in nature. If urgent, it may be addressed in the field or the shop. (The strategy for Repair Management should take into account the estimated duration of the repairs in order to make the determination as to whether repairs are performed in the field or the shop. Repairs requiring more than 30 minutes to an hour to perform are typically thought to be beyond the scope of the field service team.)

  The next step is to determine if the problem is component-related. If it is not, the repair falls under the guidelines defined by the Repair Management process. If it is component related, the guidelines defined in the Component Management strategy are applied, e.g. repair/replace criteria, component repair/life history, and the component replacement plan. If the repair option is decided, once again the repair falls under the guidelines defined by the Repair Management process. If replacement is deemed necessary, the machine is either driven or transported to the main shop for service.

When the machine arrives in the main shop, the workorder is transferred from the field to the shop where the supervisor assigns manpower. Technicians will first verify that the diagnostic decisions taken in the field are accurate and, if found to be incorrect, the proposed action must be revised accordingly.

Assuming the analysis and decisions are correct, Planning is advised and at that point they will attempt to treat the situation much the same as if it were a planned component replacement. Planning will define and procure resources including the component, R&R parts kit, people, facilities, tooling, equipment, and information just as though the replacement was planned. In addition, Planning should review the backlog repair list to determine if the parts and resources are available and decide if the "window of opportunity" presented by stoppage is sufficient to allow backlogged repairs to be made in parallel with component replacement.

Once the resources have been identified and obtained, the plan is in place and ready for execution. This execution takes place under the guidelines defined by the Repair Management process. The execution phase should include not only the task itself but also the commissioning of the "new" component as well as protection, handling, and transportation of the used component for return. The final steps are to perform a quality check of the work and to document the process via the workorder for inclusion in the machine/component history files.

- Component Replacement

Planned component replacement (PCR) is driven by the Planning and Scheduling area and executed under the procedures outlined in the Repair Management process (please see *Planning & Scheduling* and *Repair Management* strategic Best Practices). In order to be effective, component replacement activities must also draw upon resources included in the condition monitoring, parts, and human resources areas as well. However, Planning relies heavily upon the guidelines and disciplines defined by the Component Management strategy and plan.

Component Management is responsible for providing multiple inputs that aid in the development of the component replacement plan as well as its scheduling and execution. The initial component life predictions established during the "Goals & Objectives" phase of the component management strategy serve as the baseline from which actual component exchange intervals are determined. The final determination as to when a component is to be exchanged will be based upon one or more component exchange criteria. These criteria are also best defined when the strategy is developed and will require that multiple exchange parameters be monitored to determine optimum useful lives for each component.

In addition to initial life projections, component life factors that relate to equipment maintenance, application, and operation must be monitored to insure that component lives are optimized and component failures can be avoided. Acceptability criteria related to each of the equipment maintenance, application, and operation life factors must be established when the component exchange strategy is developed. They must be tracked, trended, monitored, and factored into the timing of the exchange plan. It is also necessary to define a tracking system and methodology to record and monitor each of the component life factors against initial targets. This system should include the capability to track not only actual performance relative to those targets but also to record component repair/rebuild history, movement history, failure mode, and workorder history. The database information is obtained from component condition monitoring and is managed/archived to support the component management process.

Once it has been determined that a component is due to be exchanged, Planning will need to pull information from other resources defined by the component replacement strategy. Replacement component source options include dealer rebuild, Reman, or new. Component availability and acquisition time are key factors to be considered in the decision making process. Cost is also a consideration and details such as component exchange quality and downtime need to be included in addition to the initial cost of the component to the contract. The final plan should consider and include availability of labor (manpower), facilities, tooling, installation parts kits, and special instructions associated with the exchange. Other jobs (backlogged repairs) that will take place during the "window of opportunity" provided by the component replacement need to be considered as well. Once this information has been defined and procured, the job can be sent to Scheduling for inclusion in the weekly program.

The execution phase should include not only the task itself but also the commissioning of the "new" component and protection, handling, and transportation of the used component for return. Just as was the case with the unplanned component replacement, the final steps are to perform a quality check of the work and to document the process via the workorder for inclusion in the machine/component history files.

4.0 Benefits

Implementation and execution of a detailed, fully functional Component Management process will support the following top-level goals of most mining organizations:

- ✓ Component life … component lives are optimized and may potentially be extended as a result of a Component Management strategy based upon conditional criteria.
- ✓ Maintenance costs … manpower is deployed proactively (on a planned basis) rather than reactively (crisis "management") reducing labor force requirements and costs via more efficient use of labor resources; equipment breakdowns, and resultant unplanned repairs are held to a minimum reducing both repair downtime and costs of failure-related contingent damage.
- ✓ Component reliability … component reliability is optimized resulting in a reduction in failure frequency, increased fleet availability, asset utilization, and production (tons).
- ✓ Machine productivity … equipment condition is maintained in accordance with design specifications and operating at peak performance resulting in maximized productivity (tons per hour).

Contributions of the above factors will yield minimized Cost per Ton and reduced MARC risk resulting in increased customer satisfaction and dealer profitability. Additionally, development of a Component Management strategy will aid in directing the efforts of Condition Monitoring to address the challenges of maintenance, application and operation on a site-specific basis.

Once the implementation strategy and plan have been established, the Gap Analysis Tool and process flow map for Component Management should be used to define the "as is" situation in order to identify opportunities for improvement (i.e. gaps/weak areas) in the on-site operations. Armed with this information, the site is then ready to take action to address those areas identified in the assessment exercise.

5.0 Resource Requirements

- Human Resources (on-site):

  Human resource requirements associated with the on-site execution of the Component Management plan include:

✓ Fleet Analyst (Reliability Engineer) … tracks and monitors component-related inputs for all on-site activities to insure compliance with the strategy for Component Management.

- ➢ Consolidates Condition Monitoring and component life statistics to update a component "Hot Sheet" as input to the component replacement program.
- ➢ Defines component-related Condition Monitoring routines that correlate to equipment health, application and operation.
- ➢ Reviews component-related Condition Monitoring procedures and checklists to ensure they focus on significant issues and makes revisions as required.
- ➢ Advises Planning of issues related to component health, maintenance, application and operation.

✓ Planner … develops plans including resource and time estimates to execute all on-site Component Management routines.

- ➢ Develops and maintains the medium/long-term component replacement program.
- ➢ Serves as the primary on-site communications link to external support resources, e.g. central parts organization, CRC/component rebuild facility, external contractors …
- ➢ Ensures that component-related maintenance activities, e.g. inspections, tests, adjustments, repairs, replacement, are aligned with the strategy.
- ➢ Establishes priorities for component-related maintenance activities.
- ➢ Develops the short-term plan for component-related tasks.
- ➢ Defines, organizes and procures facilities, manpower, parts, tooling, forms, checklists, etc. for component-related maintenance activities; develops these inputs into a formal plan.
- ➢ Monitors component-related maintenance and Component Management process performance metrics for compliance with goals; devises and communicates plans to address shortfalls to the Project Manager.

✓ Scheduler … translates the Component Management plan into an actionable schedule.

- ➢ Develops the short-term plan into a schedule for component-related repair and replacement activities.
- ➢ Schedules weekly component-related activities in accordance with the Component Replacement Plan/strategy and resource availability.
- ➢ Publishes/distributes the weekly schedule and obtains agreement from all appropriate areas.

- Identifies "Windows of Opportunity" for execution of backlogged repairs in conjunction with component replacement.

NOTE: The Scheduler and Fleet Analyst all work within the Planning Department and report to the Planner. Depending upon the magnitude of the operation and the scope of its responsibilities, a Planning Supervisor and multiple individuals may be required to staff each of those three key positions. Some larger operations have determined that in addition to those three positions they require specialized logistical staff whose responsibilities are to liaison with off-site resources for the purpose of procuring parts, components and external (third-party) suppliers to support efforts in planning and scheduling of component-related as well as other maintenance and repair activities. A detailed discussion of planning and scheduling process for component repair and exchange can be found in the Planning and Scheduling strategic best practice.

- ✓ <u>Operators</u> … identify component-related defects and health issues on an ongoing basis before, during and after the shift.
    - Continuously assess component condition, health and performance.
    - Document observations through pre & post-shift checklists and the machine logbook.
    - Take appropriate preventive actions in response to input from on-board warning systems.
    - Communicate component condition and performance issues to Dispatch, Field Service staff and Inspectors.

- ✓ <u>Field Inspectors</u> … participate in all activities in the field in accordance with the weekly plan.
    - Translate component-related Operator input into root-cause technical information.
    - Execute all periodic/daily field inspections.
    - Participate in the execution of pre-PM inspections.
    - Participate in the execution of pre-PCR inspections.
    - Provide accurate, complete information on inspection checklists and field documentation.
    - Generate/enter backlog repair requests.

- ✓ <u>Technicians</u> … execute all planned and unplanned maintenance activities in compliance with defined guidelines and procedures.
    - Acquire thorough knowledge and understanding of all component/systems operation and apply that knowledge in the execution of component-related maintenance tasks.
    - Be familiar with proper use and operation of all specialized tooling and instrumentation related to component diagnostics/troubleshooting, repair and replacement (R&I) activities.
    - Document results; provide accurate, complete information on checklists and forms.
    - Generate backlog repair requests for defects that cannot be addressed immediately.

- ✓ Project Manager … primary on-site individual responsible for the implementation and execution of the Component Management strategy/plan.
  - Total familiarity with customer expectations, contractual commitments and the Component Management strategy.
  - Authorizes resources to support successful implementation of the Component Management plan.
  - Authorizes revisions to Component Management strategy and procedures.
  - Authorizes modifications to the component exchange program.

- Human Resources (off-site):

- ✓ Component Specialist … primary off-site contact individual responsible for interacting with mining operations on component support issues.
  - Receives the medium/long-term plan from multiple sites within a territory and factors those needs into the CRC/Rebuild Shop program.
  - Communicates with on-site operations relative to availability of components to satisfy component rebuild/exchange requirements defined in the short-term plan.
  - Manages transportation/return of components to be rebuilt.
  - Receives repair/life history data from on-site operations for components to be rebuilt.
  - Communicates cause of failure information to on-site mining operations.

- ✓ Parts Specialist … primary off-site contact individual responsible for interacting with mining operations on parts support issues related to components.
  - Manages Reman and new component inventory to supplement CRC/ Rebuild Shop rebuild capability.
  - Procures component R&I parts kits from Parts Department to support on-site component exchange requirements.
  - Manages transportation/return of Reman cores.
  - Receives commissioning and post-installation fluid samples from newly installed components; exception-based results communications back to sites.

- Logistical Resources:

Logistical resource requirements associated with the execution of the component repair/exchange plan include:

- ✓ Facilities
  - Bays adequately sized to meet needs of largest equipment.
  - Quantity sufficient to minimize delays.
  - Compressed air, water, & electricity supplies available.
  - Fluids handling … evacuation, delivery (filtered, metered and quick disconnects) and spillage clean-up capability.
  - Work environment … well lighted, portable lighting, doors for dust/ contamination control, quality work surface, exhaust/smoke control.

✓ Tooling

- Technician hand tools, tool boxes, and work carts.
- Special tooling … pneumatic tools, filter cutting device, oil extraction (SOS), etc..
- Testing & diagnostic instrumentation … pressure/temperature/speed packages, particle counter, etc.

✓ Equipment

- Personal safety equipment and first aid provisions.
- Workbenches equipped with vices.
- Workstation … writing surface w/ computer access.
- Cleaning materials … parts washing/solvent tank, shop vacuum, shop towels, rags, soap, etc.
- Filter cart ("kidney looping"), lifting devices, chains/slings, hydraulic jacks and stands, ladders and platforms, wheel chocks, ...
- Fire protection/extinguishing devices.

✓ Information

- Weekly schedule for component repair/exchange tasks.
- Work orders for component repair/exchange tasks.
- R&I guidelines, checklists and procedures specific to each component repair/exchange task.
- Service/technical literature.
- Special instructions, etc.

Significant implementation costs include logistical resources required to execute the Component Management plan, e.g. facilities, tooling and equipment, as well as protective/PCR component inventory acquisition and carrying costs.

6.0 Supporting Attachments

- Cat Global Mining Equipment Management Process Map (Component Management)
- "*Metrics (KPI's) to Assess Process Performance*" - Microsoft Word Document

7.0 Related Best Practices

- MARC Rate Development (ref. 0609-3.3-1009)
- Getting Started with MCTS (ref. 0407-3.1-1076)
- MCTS Used to Monitor & Manage Component Life (ref. 0906-3.1-1020)
- Managing Exchange Component Inventory Reduces Costs (ref. 0107-4.5-1055)
- Ensuring Rebuild Quality through Certified Removal & Installation (ref. 1006-2.3-1026)
- Major Component Installation Kits Improve R&I Process (ref. 0107-2.3-1052)
- Recommended Shipping & Storage Stands for Major Components (ref. 0107-4.4-1053)

8.0 Acknowledgements

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